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To cite this article: Dikla Eckshtain & Scott T. Gaynor (2009) Assessing Outcome in Cognitive Behavior Therapy for Child Depression: An Illustrative Case Series, *Child & Family Behavior Therapy*, 31:2, 94-116, DOI: [10.1080/07317100902910398](https://doi.org/10.1080/07317100902910398)

To link to this article: <https://doi.org/10.1080/07317100902910398>



Published online: 06 Jun 2009.



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Assessing Outcome in Cognitive Behavior Therapy for Child Depression: An Illustrative Case Series

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Recent meta-analytic data suggest a need for ongoing evaluation of treatments for youth depression. The present article calls attention to a number of issues relevant to the empirical evaluation of if and how cognitive behavior therapy for child depression works. A case series of 6 children and a primary caregiver received treatment—individual CBT for the child and behavioral parent training involving the caregiver and caregiver–child dyad. The effects were generally promising and illustrate how selection of inclusion criteria, measures, measurement intervals, and informants can alter conclusions. These areas warrant attention in studies of child depression and are important not only in interpreting treatment outcome data but also for conducting sound clinical practice.

KEYWORDS *child depression, cognitive-behavior therapy, multi-informant, multi-method*

A recent meta-analysis of 35 psychosocial treatment studies for youth depression revealed only modest effect sizes for therapy compared to no treatment ($ES = .41$) and active control conditions ($ES = .24$; Weisz, McCarty, & Valeri, 2006). The majority of the treatments included in the meta-analysis could be considered cognitive-behavioral (CBT) in nature. This is noteworthy because CBT has generally been considered the psychosocial treatment of choice for youth depression (see Chorpita et al., 2002), but the mean effects reported by Weisz and colleagues were substantially lower than in prior

Received 20 September 2007; revised 9 May 2008; accepted 12 May 2008.

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reviews (e.g., Lewinsohn & Clarke, 1999; Reinecke, Ryan, & DuBois, 1998). These data illustrate the need for ongoing evaluation of treatments for youth depression, which is especially true for preadolescents where there is a paucity of outcome research. For instance, in their meta-analysis, Weisz et al. identified only seven studies with pre-teens compared to 21 with adolescents. In addition, while several large-scale randomized trials have been conducted with depressed adolescents (e.g., Brent et al., 1997; Clarke, Rohde, Lewinsohn, Hops, & Seeley, 1999; Rohde, Clarke, Mace, Jorgensen, & Seeley, 2004; The Treatment for Adolescents with Depression Study Team [TADS], 2004), nothing comparable has been done with children. A greater focus on adolescents is sensible in that the rates of depression rise dramatically in the teen years. However, earlier-onset depression is clinically significant and might indicate a unique and possibly more severe course (Birmaher et al., 1996; Kovacs, 1996); thus, warranting research focus in its own right.

In interpreting the modest clinical outcomes with depressed youth, a number of issues warrant consideration. One important area is how depression is measured, which has implications for the assessment of treatment outcome directly, but also indirectly in determining which children are eligible to be included in a study. To illustrate, consider the findings of Asarnow, Scott, and Mintz (2002) that 10 twice weekly sessions of CBT was more efficacious than a wait list condition for 4th–6th graders ($N = 23$). Asarnow and colleagues (2002) used a Child Depression Inventory (CDI; Kovacs, 1992) score of greater than or equal to 8 to determine inclusion. The minimum inclusion criterion, a score of 8, may appear quite modest given Kovacs' report of a mean of 9.5 for adjustment disorder with depressed mood and 11.8 for those with major depressive disorder. Indeed, extrapolating from the CDI, Asarnow and colleagues estimated that only 48% of the sample could tentatively be considered to have met DSM criteria for major depression.

However, it is plausible that even if the CDI scores were relatively low a greater proportion of the sample may have met DSM criteria for major depression than the CDI data would suggest. In a study showing a moderate benefit for fluoxetine compared to placebo ($ES = .51$; Emslie et al., 2002), children were required to meet DSM criteria and show at least moderate depressive symptoms according to the clinician-administered, semistructured Children's Depression Rating Scale-Revised (Poznanski & Mokros, 1996), which was also the primary dependent variable. Despite average CDRS-R scores that were in the mid-50s, well above the authors' cutoff for inclusion of 40, the "mean baseline CDI scores were below the cut point for depression" (Emslie et al., 2002, p. 1210). In other words, Emslie et al. found that a group of children required to meet DSM criteria and display moderate symptoms according to a clinician administered interview, failed to show elevations on the CDI. Given that clinician interviews are often considered the "gold standard," it may be that Asarnow and colleagues' sample was more severe than their CDI scores suggested. Alternatively, given the lack of corresponding

self-report data, it might be argued that Emslie et al.'s sample should be considered more modest in their symptoms than the interview suggested.

The interpretive problem, in part, is one of a lack of additional information to help qualify conclusions. Asarnow et al. (2002) did not report the pretreatment group means on the CDI and did not collect additional interviewer-administered or parent-rated symptom measures. As such, the CDI served as the sole measure for study inclusion and evaluation of depressive symptom outcome, complicating the interpretation of results. The same argument can be made with respect to Emslie and colleagues' (2002) failure to present the self-report data in their study. In a previous study by that group (i.e., Emslie et al., 1997), significant differences were observed on interviewer measures, but no differences between fluoxetine and placebo were found on parent or child self-report measures—a result which has been found in most studies using Selective Serotonin Reuptake Inhibitors (SSRIs) to treat child depression. As such, the efficacy data for SSRIs come almost exclusively from clinician-administered interviews (Jureidini et al., 2004). If children and parents do not see the problem or the effects of an intervention in the same way or to the same degree that a clinician does, this is important information.

One of the largest outcome studies with elementary-age children compared eight sessions of Primary and Secondary Control Enhancement Training (PASCET, a CBT intervention) to no treatment (Weisz, Thurber, Sweeney, Proffitt, & LeGagnoux, 1997). The 48 3rd–6th grade participants met inclusion criteria of either a CDI ≥ 11 or a CDRS-R (child informant) ≥ 34 . CBT resulted in significantly greater reductions in depressive symptoms on the CDI (a decrease from 18.6 to 7.1 in CBT compared to 17.8 to 11.8 among controls, $ES = .45$). In addition, a greater amount of change was observed on the CDRS-R for the CBT group (i.e., a decrease from 45.3 to 33.2) compared to controls (a decrease from 38.4 to 34.9). The conclusions from this study were consistent across self-report and interview measures. Data from other informants were not taken.

The potential for disagreement across measurement methods leads to the basic recommendation to take and report multiple measures using different methods (e.g., self-report and clinical interview). Weisz et al. (2006) found that on average two depression measures were collected in youth depression studies; but, these were averaged in their analyses to produce a single effect size for each study. Disagreement across measures, while possible, is only part of the interpretive dilemma. Another is disagreement across informants. A large-scale meta-analytic review revealed correlations between parent, teacher, child, and clinician reports on dimensional measures of child (internalizing and externalizing) behavior that averaged about $r = .25$ (Achenbach, McConaughy, & Howell, 1987). A review focusing on semistructured and structured diagnostic interviews also documented poor concordance across informants, particularly parents and children (Grills &

Ollendick, 2002). For instance, in a large study using the highly structured NIMH Diagnostic Interview Schedule for Children (DISC; Shaffer et al., 1996), it was found that parent-child agreement for depressive disorders produced Kappa coefficients near zero. Parent-adolescent agreement was a bit better, but still poor, $\kappa \sim .20$ (Jensen et al., 1999). Follow-up interviews by clinicians of cases identified only by the parent or only by the child resulted in “confirmation” of the depressive disorder diagnosis in only 41% and 30% of the cases, respectively, revealing substantial clinician-parent and clinician-child disagreement. With respect to parent-child disagreement, a further complicating possibility comes from data suggesting a modest correlation between maternal internalizing symptoms and their reports of their children’s symptoms, which could potentially contribute to the discrepancy (Briggs-Gowan, Carter, & Schwab-Stone, 1996).

The lack of concordance among measures and informants is relevant to and can complicate interpretation of clinical outcome data. DeCuyper, Timbremont, Braet, De Backer, and Wullaert (2004) compared 16 weekly 60-minute sessions of CBT to no treatment for children 9–13 with symptoms of depression. The CDI decreased 2.6 points (from its baseline of 12.7) for the CBT group ($n = 9$) and 3.5 points (from its baseline of 15.3) for the waitlist controls ($n = 11$), suggesting no acute treatment effect for CBT according to child self-report. However, according to parent-report on the Child Behavior Checklist-Internalizing scale (CBCL-I), there was a change from baseline (64.0) of 6.2 points for the CBT group, while in the control group the change from baseline (67.5) was less than a point. The posttreatment CBCL-I effect size of 0.99 (using Hedges correction for small samples) suggests a large effect favoring CBT. (When the effect size is adjusted by subtracting the pretreatment effect size from the posttreatment effect size, the result was a medium effect size of .55 favoring CBT.) As such, conclusions about the acute efficacy of CBT differed by informant. The Weisz et al. (2006) meta-analysis data further support this contention. That is, their compilation of studies produced only 6/35 that included both parent and youth reports of depressive symptoms. Despite the small number of studies available for analysis a significant difference was apparent, with larger effects for youth-reported compared to parent-reported symptom change. Thus, despite the DeCuyper et al. (2004) findings, the treatment efficacy data come mainly from the reports of youth.

In the preceding pages it has been suggested that interpreting the results of child depression studies is complicated by the use of single measures, disagreement between measures when more than one is used, and disagreement between informants completing measures on the same construct (i.e., depressive symptoms). One additional complication is the lack of data related to the time course of change in child depression studies. In CBT for adult depression a great deal of change appears to occur very early in treatment and may be due to nonspecific factors (Ilardi & Craighead,

1994). With depressed adolescents, time course data suggest that across treatment approaches about 25% experience notable improvements in depressive symptoms prior to the start of treatment and over 25% experience a large “sudden gain” early in the course of treatment (Gaynor et al., 2003). Determining the trajectory and stability of symptoms in the baseline period preceding treatment and the pattern of response during treatment can alter how global pretreatment to posttreatment changes are interpreted. The time course of change with preadolescents has received almost no attention in the clinical outcome literature.

The present study illustrates the importance of multi-measure, multi-informant repeated measures assessment. Six school-age children were provided individual CBT (via the PASCET manual) plus Caregiver-Child Relationship Enhancement Training (C-CRET; behavioral parent training skills added by the authors to augment the skills taught in the PASCET manual). Because of the potentially significant discrepancies between measures and informants, a multi-measure, multi-informant approach to evaluating change in depressive symptoms was used. Specifically, dimensional questionnaire data and categorical interview data from both the child and his/her primary caregiver were obtained and used in the evaluation of outcome. In addition, the short form of the CDI was completed at each session providing information on the time course of change in depressive symptoms from the perspective of the child.

METHOD

Participants

Six elementary school children, 5 females and 1 male ($M=10$; $SD=.75$), recruited/referred from Kalamazoo Center for Medical Studies/Michigan State University, Department of Pediatrics participated (see Table 1). They were required to report at least mild-to-moderate symptoms of depression as assessed by a CDI of ≥ 8 (as in Asarnow et al., 2002) *and* a report of at least an intermediate major depressive episode, by either parent or child report, on the DISC (Shaffer, Fisher, Lucas, Dulcan, & Schwab-Stone, 2000). Out of the 6 children, 5 lived with only one primary caregiver. Further, according to the caregiver, 5 had a first degree relative with a history of depression, who in four cases was the biological mother who also served as the study informant. One child took medications but was eligible to participate because he was on the same medication and dose for more than 2 months prior to enrollment, met entry criteria, and his medication doses did not change during the time of treatment. Two other children had received psychotropic medicine previously, but were not currently medicated. Exclusion criteria included primary symptoms of extreme Conduct Disorder, Obsessive-Compulsive Disorder, schizophrenia-spectrum, autism-spectrum disorders,

TABLE 1 Demographic and Clinical Characteristics

Participant	Sex	Age	Grade	Ethnicity	Live with 2 caregivers	Psychiatric Meds (Now/Ever)	Family History of Depression	Informant	Informant Mental Health History	Individual sessions	Caregiver sessions
1	Female	10	5	EA	Yes	No/No	No 1st degree (Mother)	Biological Mother	Depression	13	7
2	Female	9	4	EA/ West Indian	No	No/No	Yes 1st degree (Mother)	Biological Mother	Depression, Anxiety, Inattention	13	7
3	Female	10	4	EA/AA/NA	No	No/Yes	No 1st degree (Mother)	Biological Mother	Depression, Inattention, Hyperactivity, Aggression	13	7
4	Female	9	3	EA	No	No/Yes	Yes 1st degree (Mother)	Biological Mother	Depression, Anxiety, Aggression	7	6
5	Female	10	4	AA	No	No/No	No 1st degree (Father)	Biological Mother	None	12	7
6	Male	11	6	EA	No	Yes	Yes No	Non-biological Grandmother	None	13	7

Note: AA = African-American; EA = Euro-American; NA = Native-American.

or report of serious acute suicidality. Two children did not meet entry criteria on the CDI and a third child met exclusion criteria due to Asperger's Disorder.

Design

All participants received PASCET+C-CRET. No concurrent waitlist group was utilized. As such, this study was an uncontrolled open clinical trial. To provide multiple assessments of outcome, diagnostic interviews, parent checklists, and child self-report measures were administered at pre-treatment, midtreatment, posttreatment, and 1-month follow-up. In addition, child-reported depressive symptoms were measured at three baseline assessments, prior to each treatment session, at 1-week posttreatment and follow-up. These session-by-session data yield a series of single-case A/B designs where A was baseline and B was PASCET+C-CRET.

Intervention

The individual treatment was based on the PASCET manual (Weisz, Moore, & Southam-Gerow, 1999). The caregiver/caregiver-child sessions included the psycho-educational meetings routinely used in PASCET plus 4 additional sessions of C-CRET (Eckshtain & Gaynor, 2003) to augment the PASCET skills. The treatment was broken into two phases separated by a mid-treatment assessment. At the maximum, the intervention consisted of 13 individual meetings with the child, and 7 caregiver/caregiver-child meetings (see Table 2). Each session lasted 75–90 minutes.

The PASCET program is a structured intervention for children ages 8–15. The treatment is based on a two-process model of control. According to this model, primary control involves enhancing reward or reducing punishment by making, to the extent possible, objective conditions conform to the individual child. Secondary control involves enhancing receipt of rewards or reducing punishment by adjusting oneself to fit environmental conditions so as to influence their subjective impact without altering the events themselves. Primary control is applied to distressing conditions that are modifiable and secondary control to those conditions that are not (Weisz et al., 1999; Weisz, Southam-Gerow, Gordis, & Connor-Smith, 2003).

PASCET caregiver sessions served the general functions of keeping the caregivers informed, allowing for feedback regarding how the child is doing outside the treatment setting, and encouraging the caregivers to assist their children in generalizing the skills they worked on in the individual sessions to the home environment. The added C-CRET sessions focused on caregiver skills, including increasing the amount of positive time spent together between the caregivers and child, positive reinforcement, noncontingent praise, enhancing communication and encouraging positive communication, familial problem solving, and increasing the child's

TABLE 2 Sequence and Contents of PASCET and C-CRET Sessions

Week	Child session	Caregiver session	Emphasis/Skill
1	1	1	Orientation and psychoeducation Introduction and problem solving
2	2	2	Special time and positive behavior management Activity scheduling
3	3	3	Activity scheduling Learning to relax
4	4		Presenting a positive self
5	5		Developing talents and skills
6		4	Midtreatment assessment Case formulation review
6	6		Think positive—no negative thinking allowed
7		5	Communication training; positive communication
8	7		Good things to do when bad things happen
	8	6	Family problem solving
9	9		Combining primary and secondary control
	10		Using skills in my everyday life
			Review of case formulation and preparing for individually tailored sessions
10	11		Individually tailored session focused on the most relevant skills for the particular child
	12		Individually tailored session focused on the most relevant skills for the particular child
11	13		Review of progress
		7	Review of progress
12			Posttreatment assessment

engagement in positive activities. Table 2 provides a summary of each session and its major focus.

Treatment Integrity

All treatment sessions were conducted by the first author (DE), a doctoral student in clinical psychology, who had completed graduate courses in child, adult, and family behavior therapy. She had 2 years of clinical experience working with children conducting both assessments and cognitive-behaviorally focused treatment. In addition, she received weekly supervision from the second author (SG), a licensed clinical psychologist with research and clinical experience using CBT with youth. To promote integrity, the PASCET manual was studied in great detail prior to treatment implementation and in developing the C-CRET sessions to accompany the PASCET manual. In addition, prior to conducting each individual and caregiver meeting for the first time, the therapist role-played the session with a research assistant.

To assess treatment adherence, after every session the therapist completed a Therapist Record Form. The PASCET forms were taken directly from the PASCET manual developed by Weisz et al. (1999), while the C-CRET rating forms were developed for the current treatment using the same general

format as used with the PASCET. As a check, all sessions were videotaped and 37% of them were reviewed by graduate student coders in a doctoral program in clinical psychology. The coders watched the tapes and completed the Therapist Record Form. The coded sessions were chosen randomly with the restriction that for each participant one session had to come from the first four individual sessions, one from sessions 5–8, one from sessions 9–13, and one caregiver session from the four C-CRET sessions. Comparison between the therapist and the three coders indicated 85% agreement, 11% disagreement, and 4% uncodable items, yielding $\kappa = .72$ ($n = 284$, when uncodable instances were coded as disagreement) and $\kappa = .76$ ($n = 274$, when uncodable instances were not included). The percent agreement and Kappa coefficients suggested good to very good agreement between the therapist and the coders (Cicchetti, 1994; Watkins & Pacheco, 2000). With 1.0 indicating perfect adherence, the average therapist coding was .97 ($SD = .07$) and the average coders rating was .88 ($SD = .12$), indicating that the therapist implemented treatment with fidelity to the manuals.

Primary Measures

The *Computerized NIMH DISC* (Shaffer et al., 2000) was used to assess psychopathology across the major diagnostic categories of the DSM-IV. The modules for major depression, dysthymia, bipolar disorder, generalized anxiety, social anxiety, conduct disorder, and Attention-Deficit/Hyperactivity Disorder (ADHD) were administered at pretreatment, midtreatment, post-treatment, and follow-up. The DISC is considered a reliable tool with moderate to good validity in assessing child psychopathology (Schwab-Stone et al., 1996; Shaffer et al., 1996). The DISC was administered to the children (DISC-c) and to the caregivers regarding their children (DISC-p). For this study, the clinical cutoff on the DISC was meeting DSM criteria for a major depressive episode of at least intermediate severity.

The CDI (Kovacs, 1992) is a commonly used self-report measure of depression for children that consists of 27 items assessing the presence and severity of symptoms during the previous 2 weeks. The CDI has acceptable internal consistency and test-retest reliability (Finch, Saylor, Edwards, & McIntosh, 1987; Kovacs, 1992). The CDI was administered at pretreatment, midtreatment, posttreatment, and follow-up. The 10-item CDI-Short Form was administered to the children at the remaining baseline sessions and prior to each treatment session. Kovacs (1992) reported a mean of 9.5 for adjustment disorder with depressed mood and 11.8 for those with major depressive disorder. The latter was considered the clinical cutoff in this study.

The *Child Behavior Checklist-Parent Rating Form* (CBCL; Achenbach & Rescorla, 2001) is composed of 118 items assessing internalizing (CBCL-I) and externalizing (CBCL-E) behaviors of the child. The CBCL has acceptable test-retest reliability and content, construct, and criterion-related validity

(Achenbach & Rescorla). Consistent with standard use of the CBCL, *T* scores below 60 were considered in the normative range, scores from 60–63 in the borderline range, and higher scores the clinical range.

Secondary Measures

The *Cognitive Triad Inventory for Children* (CTI-C; Kaslow, Stark, Printz, Livingston, & Ling-Tsai, 1992) is a self-report measure for children consisting of 36 items assessing their view of the self, world, and future. The CTI has demonstrated acceptable internal consistency and concurrent validity, with depressed children (in grades 4–7) averaging 39.5 (11.3) and control children 63.3 (5.1).

The *Children's Negative Cognitive Error Questionnaire* (CNCEQ; Leitenberg, Yost, & Carroll-Wilson, 1986) is a 24-item self-report measure assessing cognitive errors (i.e., overgeneralization, catastrophizing, personalizing, and selective abstraction). The CNCEQ is reported to have test-retest reliability of .65 for the total score. Average scores of 70.3 for depressed children (in grades 4, 6, and 8) and 42.1 for nondepressed children were reported.

The *Matson Evaluation of Social Skills for Youngsters* (MESSY; Matson, Rotatori, & Helsel, 1983) is a self-report measure of social skills, which has demonstrated acceptable internal consistency and test-retest and split-half reliability (Matson, Esveldt-Dawson, & Kazdin, 1983; Matson, 1994). The items making up the *Appropriate Social Skills*, *Inappropriate Assertiveness*, and *Impulsivity* scales were completed by the child (MESSY-c). The later two were subtracted from the former, such that higher scores suggested better social skills. Normative averages for 9- to 12-year-olds according to this scoring method are 46 for girls and 37 for boys (Matson, 1994). Caregivers were given the teacher-report version of the MESSY (MESSY-p). This version consists of 64 items and has been used with caregivers in other research yielding a normative mean (SD) of 139 (25; Bell-Dolan & Allan, 1998).

The *Self-Report Measure of Family Functioning* (SRMFF; Bloom, 1985) is a 75-item measure completed by the primary caregiver. The SRMFF has been reported to have adequate psychometric properties. In addition, young adults who reported coming from disrupted families described their families differently ($M = 184$) from young adults who reported coming from non-disrupted families ($M = 196$; Bloom). A family mental health history was obtained as part of a demographic questionnaire. The caregiver informant responded to the question “Does anyone in the child’s immediate or extended family have the following illnesses or problems?” by indicating the relevant family member next to each of the applicable 24 possible problem areas listed.

RESULTS

Table 3 presents the data from the primary measures collected at pretreatment, midtreatment, and posttreatment and displays the total number of

TABLE 3 Measures of Depressive Symptoms

Part	Pretreatment					Midtreatment					Posttreatment							
	Child			Caregiver		Child			Caregiver		Child			Caregiver				
	CDI	DISC-c		CBCL-I	DISC-p	Cutoff	CDI	DISC-c		CBCL-I	DISC-p	Cutoff	CDI	DISC-c		CBCL-I	DISC-p	Cutoff
1	19	MDE:I		68	MDE:I	4	19	MDE:I		71	No dx	3	8	No dx		70	No dx	1
2	23	No dx		76	MDE:I	3	4	No dx		88	MDE:I	2	5	No dx		85	MDE:I	2
3	8	No dx		71	MDE:I	2	5	No dx		63	No dx	0	0	No dx		59	No dx	0
4	15	No dx		82	MDE:I	3	9	No dx		67	No dx	1	–	–		–	–	–
5	9	No dx		58	MDE:I	1	0	No dx		58	No dx	0	0	No dx		52	No dx	0
6	9	No dx		64	MDE:I	2	13	No dx		68	MDE:I	3	7	No dx		55	No dx	0
Cutoff	3	1		5	6	15	2	1		4	2	9	0	0		2	1	3

Note: The DISC columns only include diagnostic information for depression. For P2, the posttreatment DISC-p also yielded diagnoses of SAD and ODD. For P6, the pretreatment and midtreatment DISC-c interviews yielded diagnoses of SAD and ADHD, while the DISC-p interviews yielded additional diagnoses of ADHD and ODD at pretreatment and midtreatment and ODD at posttreatment.

CDI = Child Depression Inventory.

DISC = Diagnostic Interview Schedule for Children: DISC-c and DISC-p refer to whether the child or parent was the informant.

CBCL-I = Child Behavior Checklist-Internalizing Symptom Scale.

Cutoff = The total number of measures (out of four) or cases (out of six) exceeding clinical cutoffs at that assessment point.

MDE:I = Major Depressive Disorder: Intermediate Impairment.

cases, and measures within cases, that exceeded clinical cutoffs at each assessment point. Figure 1 provides the CDI time course data and Table 4 the data from the secondary measures. With respect to Table 4, the focus is on measures for which pretreatment scores appeared to reasonably indicate dysfunction according to existing normative data. These instances are “boxed” in Table 4 for easier identification.

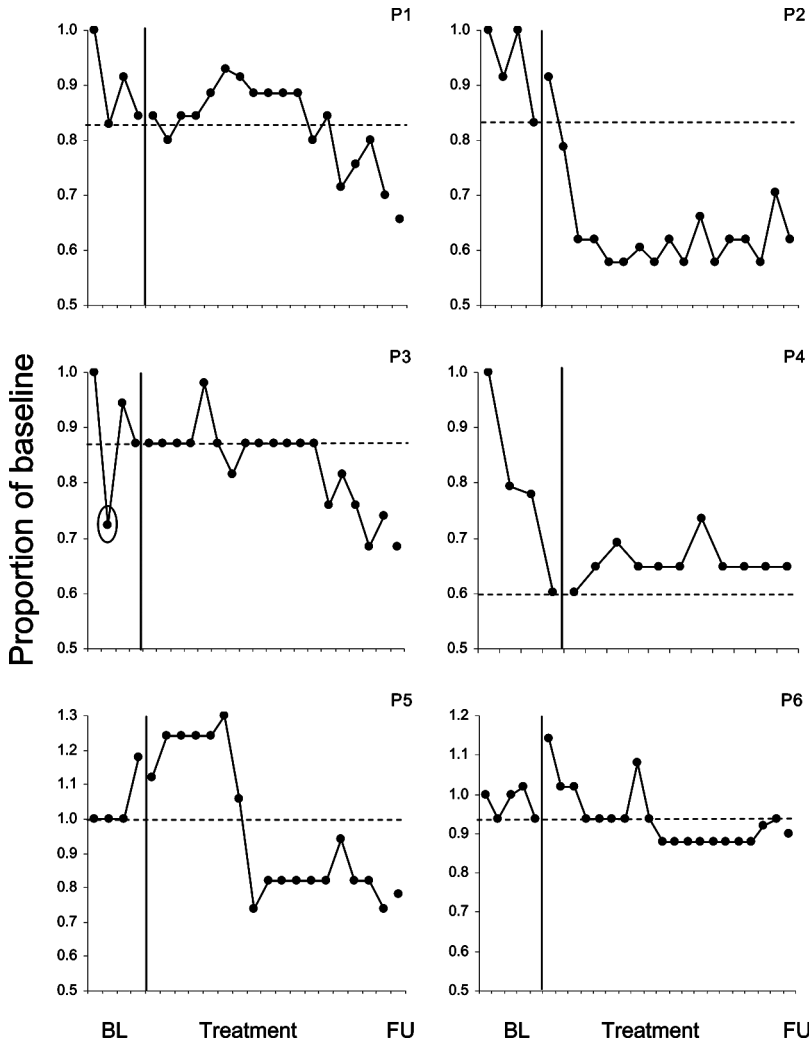


FIGURE 1 Individual repeated measures data across baseline (BL), treatment, and 1-month follow-up (FU) phases presented as a proportion of the baseline Child Depression Inventory *T* score. *T* scores were used to equate the data from the Child Depression Inventory and Child Depression Inventory–Short Form. The dashed, horizontal line represents the level of the lowest baseline score (with the exception of P3 where one baseline session, which is circled, was unknowingly scheduled on her birthday).

TABLE 4 Secondary Measures

Variable	Participant 1			Participant 2			Participant 3			Participant 4			Participant 5			Participant 6		
	Pre	Mid	Post	Pre	Mid	Post	Pre	Mid	Post	Pre	Mid	Post	Pre	Mid	Post	Pre	Mid	Post
CTI	56	51	57	34	56	55	65	67	72	54	70	–	71	68	58	50	66	69
CNCEQ	37	47	45	59	43	30	37	40	32	41	45	–	30	26	24	70	60	24
MESSY-c	54	36	40	46	58	65	82	75	67	90	87	–	78	74	88	46	80	87
CBCL-E	53	53	46	74	77	77	63	59	59	75	64	–	42	37	32	66	59	56
MESSY-p	144	152	141	216	216	217	149	136	130	207	165	–	122	105	108	148	150	114
SRMFF	190	204	206	210	213	217	184	188	199	211	198	–	173	208	203	206	206	211

Note. CTI: 39.50 (11.30) for depressed children, 63.3 (5.1) for control children (grades 4 to 7; Kaslow et al., 1992).

CNCEQ: 70.3 for depressed children, 42.1 for nondepressed children (grades 4, 6, and 8; Leitenberg et al., 1986).

MESSY-c: 65.2 is normative data for girls, 78.3 for boys (Spence & Liddle, 1990).

MESSY-p: 138.6 (25.1) is normative data from parents (Bell-Dolan & Allan, 1998).

CBCL: *T* score of 64 or greater is considered in the clinical range (Achenbach & Rescorla 2001).

SRMFF: 184 (clinical), 196 (control), based on retrospective report of young adults from disrupted (clinical) and nondisrupted (control) families (Bloom, 1985).

As is apparent in Table 3, P1 exceeded clinical cutoffs on all 4 measures at pretreatment. At midtreatment she no longer met criteria for a major depressive episode on the DISC-p, but did continue to exceed clinical cutoffs on the other 3 measures. However, by posttreatment she only exceeded cutoffs on the CBCL-I. These results were maintained at 1-month follow-up (i.e., CDI = 6, no DISC depression, CBCL-I = 74). The time series data (Figure 1) suggest relatively stable symptoms across baseline, which, like the DISC-c, did not reliably decrease until the second half of the treatment where a steady decline was noted with the lowest point at the posttreatment assessment. On the secondary measures, the only potential elevation was on the SRMFF, which was midway between the clinical and control sample averages at pretreatment (see Table 4). However, family functioning improved by midtreatment and remained improved at posttreatment. Thus, P1 demonstrated depressive symptom change on both the interview and self-report measures for which she was the respondent and on 1/2 measures for which her primary caregiver was the informant. The primary caretaker identified the changes in depression and family functioning as occurring within the first half of the treatment, while the child noted the most change during the second half of the treatment.

P2 exceeded clinical cutoffs on 3 measures at pretreatment, with the exception being the DISC-c. She showed a dramatic decrease on the CDI by midtreatment, which was maintained at posttreatment, decreasing the number of clinical cutoffs met from 3 to 2. The CDI time course data showed a modest dip in symptoms at the final baseline point, which rebounded to the previous level prior to session 1. However, by the start of session 3 she reported a dramatic decline in depressive symptoms, which was maintained through the remainder of treatment. Thus, according to her self-report P2 showed significant change early in the course of treatment that was maintained. That said, no depression change occurred according to maternal report. The same pattern was observed on the secondary measures. That is, according to maternal report P2 was also demonstrating significant externalizing behavior and social skills problems at all assessments. P2 did not report problematic social skills, but did report a depressotypic cognitive style at pretreatment, which was substantially improved by midtreatment (CTI & CNCEQ) and showed additional improvement at posttreatment (CNCEQ). Thus, according to child self-report substantial changes in depressive symptoms and depressotypic thinking occurred early in the course of treatment, while no changes were noted by maternal report on any dimensional measures or the diagnostic interview. P2 did not participate in the follow-up assessment.

P3 began treatment above the clinical cutoff on both parent-rated measures, but neither child self-report measure. By midtreatment P3 no longer exceeded clinical cutoffs on the DISC-p or CBCL-I. Maintenance or further improvement was observed at posttreatment and at 1-month

follow-up where she also failed to meet any clinical cutoffs: no DISC-c or DISC-p depression diagnosis, CDI = 0, and CBCL-I = 54. The time course data suggest initially mild depressive symptoms that dropped precipitously at the second baseline session (which, unbeknownst to the therapist when the session was scheduled, happened to be her birthday) and returned to mild levels during the last two baseline sessions. These mild symptoms then remained stable across the initial two thirds of treatment, with a decline in the final five sessions to a low level. On the secondary measures P3's mother reported a small decrement in externalizing (from the borderline to nonclinical range) from pretreatment to midtreatment, and a substantial improvement in family functioning during the second half of treatment. In sum, both P3 and her mother identified a change in depressive symptoms. P3's mother identified the most change in the first half of treatment with ongoing improvement in the second half. P3's self-reported change occurred primarily in the second half, the time frame in which her mother reported improvements in family functioning.

P4 terminated treatment after 7 individual sessions and 6 parent sessions due to family difficulties, including living far away, transportation problems, her mother's emotional and physical health, her father's health, and some disagreement between the parents on the need for ongoing treatment. As such, data are only available for P4 through midtreatment. At pretreatment P4 met 3/4 clinical cutoffs. The exception was the DISC-c. At midtreatment she showed decreased symptoms sufficient to move from meeting 3 cutoffs to 1. Despite a CBCL-I score of 67, which continued to exceed the cutoff, a notable decrease was observed from the baseline score of 82. A similar decrement was noted on the CBCL-E as was an improvement in parent-rated social skills. P4's repeated measures data suggest that she experienced substantial change in symptoms during the baseline period before the onset of treatment. Thus, the self-reported decrease in depressive symptoms does not appear attributable to the treatment.

P5 met pretreatment clinical cutoffs on only 1 measure, the DISC-p. By midtreatment P5 no longer met criteria for a depressive episode on the DISC-p, a result which was maintained at posttreatment. The time course data (see Figure 1) suggest that P5 reported mild depressive symptoms during baseline that were modestly worsening at the last baseline measurement and during the initial portion of treatment. However, session-by-session data suggest that about 6–8 sessions into treatment P5 experienced a relatively abrupt change in depressive symptoms to a low level that was maintained throughout treatment. Thus, the DISC-p and CDI data suggest depression change was concentrated during the latter portion of the first half of treatment. An improvement in family functioning was also reported during this time period. However, dimensional change on the CBCL-I was only observed from midtreatment to posttreatment. No clinical cutoffs were exceeded at the 1-month follow-up, CDI = 1, CBCL-I = 48.

At baseline P6 met clinical cutoffs on both caregiver-rated measures. A slight worsening was observed at midtreatment where he also exceeded the cutoff on the CDI in addition to the CBCL-I and DISC-p. However, significant improvement occurred by posttreatment, as P6 no longer exceeded clinical cutoffs on any depression measure (but did still meet DISC-p criteria for ODD). At 1-month follow-up these gains were maintained: no DISC depressive disorder diagnosis, CDI = 5, and CBCL-I = 53. The time course CDI data suggest only a small change during the course of treatment. Two modest “spikes” in symptoms were observed where his CDI increased. At the beginning of the second half of treatment his CDI scores decreased to their lowest level, which was outside the baseline range. This decrease was maintained for eight assessments before slightly increasing at posttreatment and then decreasing by follow-up. Thus, for both P6 and his caregiver, the changes in depression were most apparent in the second half of treatment. Interestingly, at midtreatment, and therefore before any documented symptom change in depression, P6 reported an improvement in his thoughts about himself, the world, and future and a modest decrement in cognitive errors. Improvement on the former remained stable during the second half of treatment, while the latter showed marked additional improvement.

DISCUSSION

Measurement issues may complicate the evaluation of treatment studies for child depression. Reliance on single measures, disagreement between measures (when more than one was used), disagreement between informants (competing measures on the same construct), and a lack of time course data may impact who is included in research studies and the data bearing on if and how they work. In the present study, enrollment required meeting an inclusion threshold according to either child or caregiver report on either a categorical or dimensional measure. If a single measure had been used to determine inclusion the number of participants would vary from 1/6 to 6/6 depending on the measure. Based on clinical cutoffs, only a single participant exceeded both categorical and dimensional measures across informants, with two others exceeding cutoffs on one child measure and the caregiver measures. Across informants, the greatest pretreatment disagreement was on meeting the diagnostic threshold on the DISC. All six caregivers endorsed sufficient DISC symptoms, but only one of the children concurred. Dimensionally, 3/6 children exceeded the CDI cutoff, while 5/6 were in the clinical range on the CBCL-I, again suggesting greater internalizing symptom endorsement by the caregivers. For P1–P4 the informant was her biological mother who, in each case, reported a significant personal history of mental health problems that included depression (see Table 1). As such, it is noteworthy that these informants produced the highest CBCL-I

scores. However, 3/4 of these children (P1, P2, & P3) had the highest initial self-reported CDI scores as well. Thus, the dimensional data did not suggest a dramatic mother-child discrepancy associated with the informant's mental health history. Instead, these pretreatment data are more consistent with the suggestion that a significant caregiver mental health history may be related to elevated child internalizing symptoms (Briggs-Gowan et al., 1996). Within informants, caregivers demonstrated good pretreatment reliability across the DISC-p and CBCL-I as 5/6 (83%) reports exceeded cutoffs on both. Children's agreement across measures was also generally good—three were not elevated on either the dimensional or categorical measure and one was elevated on both, suggesting agreement in 4/6 (67%) cases.

Depending on the measure and informant, the number of treatment responders ranged from 40–100%. At the conclusion of their treatment, 3/6 failed to meet any clinical cutoffs. Those who met more self- and informant-rated clinical cutoffs at pretreatment (P1, P2, & P4) met fewer at their final assessment, but did continue to meet at least one clinical cutoff at their final assessment. With respect to specific measures, on the CDI 6/6 children reported reductions in symptoms over the course of the treatment. All three who exceeded clinical cutoffs at pretreatment no longer did so at either mid-treatment or posttreatment. On the DISC-p, 5/6 children went from meeting criteria for a major depressive episode to no longer meeting criteria during the treatment period. On the CBCL-I, 4/6 showed dimensional decreases in internalizing problems, with 2/5 (who were initially above the criterion) no longer meeting the clinical threshold at posttreatment. For the three who still met the CBCL-I cutoff at his/her final assessment, the data were provided by a maternal caregiver with a history of mental health problems that included depression. These data suggest a potential role for this variable in accounting for the outcome discrepancy on the dimensional measure. However, it is important to note that two of these mothers reported a decrease on the DISC-p resulting in no depressive disorder diagnosis and that our measure of maternal mental health was an affirmative response on a family history item list. No measure of the informant's current symptoms was collected. In sum, these data are consistent with findings in the adolescent literature that pretreatment severity is related to reduced likelihood of meeting remission cutoffs (Clarke et al., 1992; Curry et al., 2006; Gaynor & Lawrence, 2002) and that the findings were generally strongest on youth-report measures (Weisz et al., 2006). Given the addition of the C-CRET sessions, it is also interesting to note that the three families with the lowest self-reported family functioning at pretreatment reported substantial improvements in this area over the course of treatment.

The current results are not from a randomized clinical trial and do not employ the strongest possible single-case strategies (e.g., multiple baseline design); thus, the generally positive outcome results must be viewed with these design limitations in mind. The purpose of this article; however, is

not to offer strong conclusions regarding efficacy but to highlight some important interpretive and measurement issues and to point out their implications for future research (single-participant and group designs) and clinical practice.

One ready recommendation is to use and report data from multiple measures on key constructs. This allows for documentation of method variance that can be used to clarify or qualify sample characteristics and outcome conclusions. Such a recommendation may seem obvious given the fairly longstanding recognition of the issue (see Kazdin, 1989) but, as reviewed in the Introduction, numerous recent studies of child depression treatment have failed to implement this practice. It is also worth remembering that what is being measured are multi-factorial constructs. Depression is a summary label applied when a group of five or more signs and symptoms are present most of the day, everyday, for a couple of weeks (American Psychiatric Association, 2000). This definition of depression allows for substantial subjectivity and variability in reporting (e.g., in deciding whether to respond in the affirmative to whether sad mood has been present “most” of the day and nearly everyday). Thus, it is not that one informant is necessarily correct (e.g., the child is depressed), while another is incorrect (e.g., the child is not depressed). Each brings his/her level of psychological distress, perspective, experiences, biases, interpretations, and imperfect recall to the reporting situation (De Los Reyes & Kazdin, 2005). Furthermore, in some cases certain behaviors of the child may be more salient and more likely to guide responding for one informant than others and, in some cases, controlling variables other than absolute accuracy may enter into the reporting (e.g., attempts to deny problems or to locate the source of problems with a particular individual).

Identification of discrepancies (or lack thereof) is not only important for evaluating research, but also for conducting treatment in clinical practice settings. Recent data suggest that in clinical settings parents and children often disagree on even the broad problem(s) for which services are sought (Yah & Weisz, 2001). Such disagreement can present quite a challenge for therapists in establishing agreed upon treatment goals (Hawley & Weisz, 2003) and may be a marker of parent-child conflict more generally (De Los Reyes & Kazdin, 2006). As such, taking measures from multiple informants is not just of informational value (to determine if inclusion/exclusion criteria are met, to get as well-rounded a view of the child/family as possible, etc.) but may point to the first treatment goal—to establish agreement on goals. The pervasive lack of agreement on target problems may be one reason that children fail to complete a full course of treatment (Nock & Ferriter, 2005) and may also reduce the effectiveness of treatment (Hawley & Weisz).

A number of variables differentiate research therapy from clinic-based care (Weisz, Donenberg, Han, & Weiss, 1995) and the outcomes achieved in clinical trials are not always replicated when treatment is applied in

clinical practice settings. Part of the problem may be due to a lack of understanding of how treatment for depressed youth works when it does (Weersing & Weisz, 2002). Taking repeated measures of relevant process (e.g., depressotypic thoughts, family relations) and outcome variables (e.g., depressive symptoms) can be important in better understanding the change process. For instance, the repeated measures data for 4/6 participants (P2, P4, P5, P6) suggest that the majority of their self-reported improvement in depressive symptoms happened over a very brief (1–2 week) interval. The timing of the change was different between the participants—P4's occurred prior to treatment, P2's early in treatment, and P5's and P6's in the middle of treatment—but the fact that the observed change was concentrated over a brief interval (and maintained thereafter) was similar. These data suggest that, for some, the change process may be abrupt rather than gradual. For the remaining two participants the majority of change on the repeated measure was concentrated in the later portion of treatment and was more gradual. That the sample was comprised of those demonstrating pretreatment gains, sudden within-treatment gains, and gradual gains are consistent with findings with depressed teens (Gaynor et al., 2003). What these patterns mean is unclear (especially since the caretaker data did not always agree with the pattern identified in the child's repeated measures), but a better understanding of them appears important in determining why change occurs.

More research is needed to determine the efficacy of CBT in the treatment of children with depressive symptoms. To most usefully interpret the findings and their clinical implications, multiple measures of key constructs, from the perspective of numerous informants, taken at repeated time intervals throughout treatment should be collected.

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